

ESP32-Powered AUDIO-VISUAL SIREN

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This sound alternator is designed to simulate the effects of a police siren, combining sound and light to create a dynamic audio-visual experience. It uses an ESP32, a small speaker, and two red LEDs to alternate between sound signals and blinking lights. Control is achieved through

code written in the Arduino IDE or MicroPython, utilising the GPIO pins of the ESP32.

The ESP32 regulates the sound frequency, enabling the speaker to mimic the distinct tones of a police siren. Simultaneously, it controls the voltage across the LEDs, causing them to alternate in illumination.

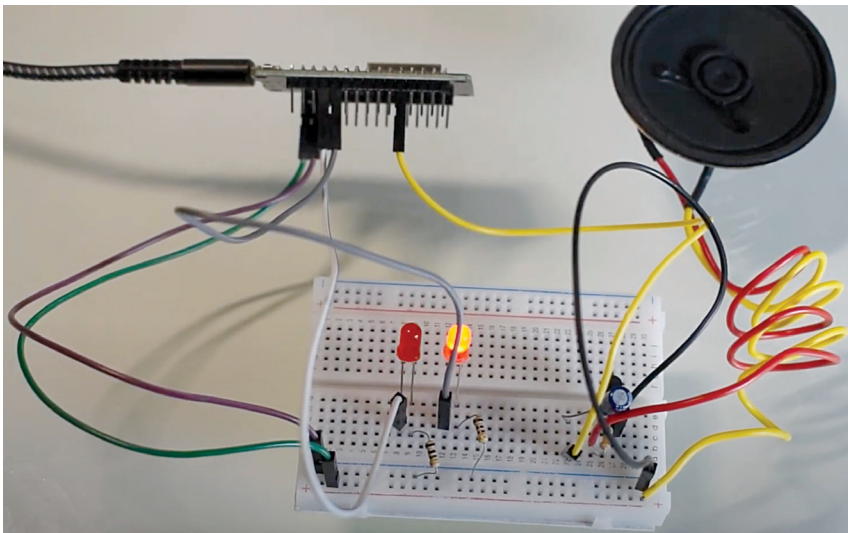


Fig. 1: Author's prototype on a breadboard

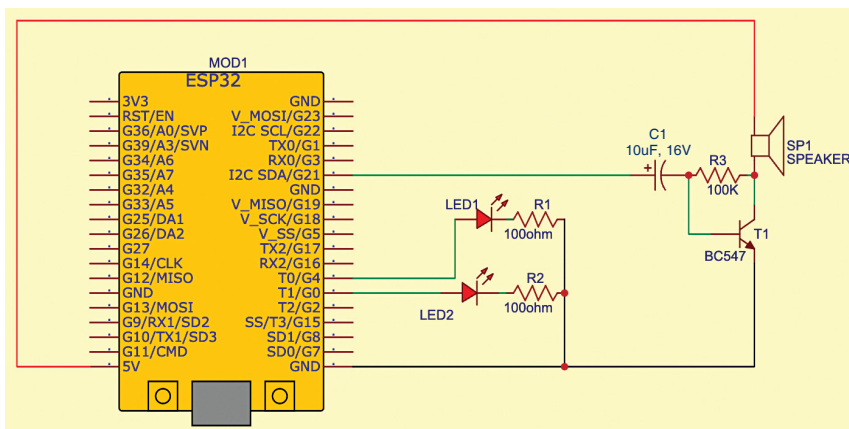


Fig. 2: Circuit diagram

Fig. 1 showcases the prototype on a breadboard, with the required components listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the sound alternator, which is built using the ESP32 board, BC547 transistor, two red LEDs, a speaker, and other components.

The circuit includes a transistor connected to a breadboard. A 100k resistor is placed across the base and collector junction of the transistor. The capacitor's negative terminal is connected to the base of the transistor, and its positive terminal is linked to pin G21 of the ESP32.

One terminal of the speaker is connected to the 5V pin of the ESP32, while its other terminal is connected to the collector of the transistor. The emitter of the transistor is grounded by connecting it to the ESP32's ground pin. To limit the current, the LEDs are also grounded via 100-ohm resistors (R1 and R2).

The anode of LED1 is connected to pin G4 of the ESP32, while the anode of LED2 is connected to pin G0. Both LEDs and the speaker are

BILL OF MATERIALS

Components	Quantity
ESP32 (MOD1)	1
0.5W speaker (SP1)	1
BC547 transistor (T1)	1
10µF, 16V capacitor (C1)	1
100kΩ resistor (R3)	1
100Ω resistor (R1, R2)	2
5mm red LEDs (LED1, LED 2)	2